L.No=5

Floatation.

→ **Fluids** - It is a substance which flow under the action of Applied force. (Liquid + Gas).

→ **Pressure** - Normal force acting per unit area on the surface of body.

$$P = \frac{F}{A} \rightarrow \frac{\text{Trust}}{A}$$

⇒ ~~Thurst~~ Trust = Normal force on the surface of body.

⇒ Unit of Pressure - $P = \frac{F}{A} = \frac{N}{m^2} \Rightarrow \boxed{\text{Pascal}}$

⇒ **Density** :- ρ (Rho)

$\rho = \frac{m}{V}$. density is defined as mass per unit volume.

- Unit - kg/m^3

☆⇒ Density of Water = $1000 kg/m^3$ at $4°C$.

☆→ ~~R.Rela~~ Relative density: $\dfrac{\text{Density of the substance}}{\text{Density of water.}}$

- Density of substance = R.D × Density of water.

Q) R.D of gold is 19.6

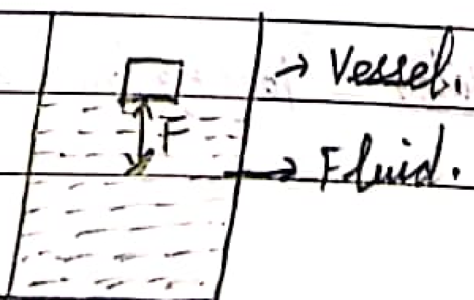
∴ Density of gold = R.D × Density of substance
$= 19.6 \times 1000\ kg/m^3$
$= 19600\ kg/m^3$.

⇒ Gold is 19.6 times hevier then equal volume of Water.

Q- R.Density of silver (Ag) is 10.3.

- Density of silver = R.D × D. of Water
$= 10.3 \times 1000\ kg\,m^{-3}$
$= 10300\ kg\,m^{-3}$

⇒ Buoyancy and Buoyant force:-



- The upward force exerted by a liquid on the surface of object, when it is immersed by a fluid is called buoyant fore/Upward Thrust.

⇒ <u>Archimede's principle</u>- When a body is immersed partially or fully it experiences a upward force which is equal to the weight of the displaced fluid by the immersed portion of the body.

$W = mg$
$\downarrow$

$T = W = mg$

$$\boxed{T = W = \rho V g}$$

Sound

⇒ Sound - Sound is a wave produced by the viberating object.

- It is a longitudinal wave.

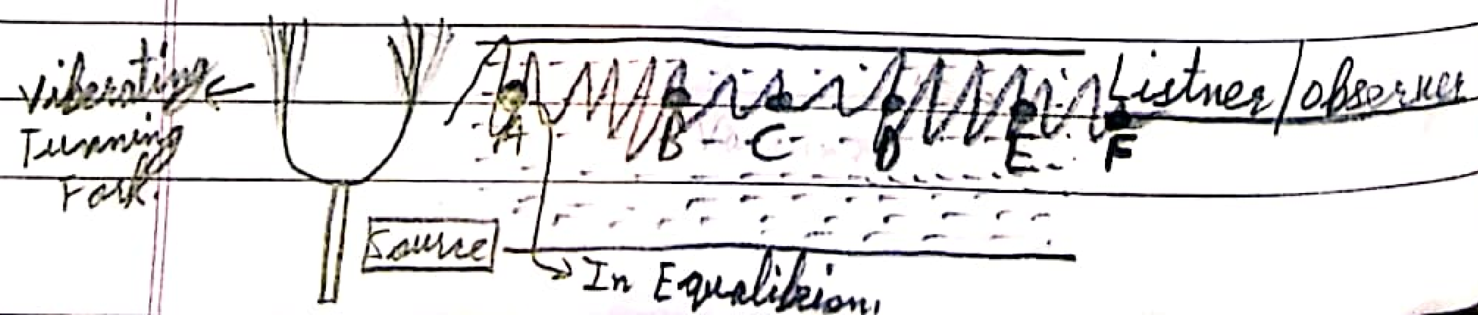
⇒ Production of sound - Sound is produced by the viberating object.

e.g Viberating turning fork.

⇒ Propagation of sound - The sound needs medium (matter or surrounding) for its propagation i.e Solid, Liquid, gas.

⇒ Sound can not propagate through Vacuum.

⇒ How it Propagates -



- The particles are viberating with greater Amplitude therefore disturbs the equalibrium of the neighbour particle.

Note:- The particle did not move/travel all the way from source to the ~~lisle~~ listener but the disturbance travel from source to listener.